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# Real-time Sign Language Translation with 3D Avatar Interaction for Inclusive Communication

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## Abstract

People with hearing impairments frequently encounter several obstacles while trying to communicate with the hearing society. To provide additional support for deaf individual for making them inclusive in all mode of communication, without using conventional video imaging. This article suggests constructing a prototype of 3D avatar techniques and finger movement tracking with animation to incorporate the avatar into video chats. This bridges the gap between deaf-mute people and hearing individuals by translating the speech into Indian sign language (ISL), where the signs are symbolized by the 3D avatar. A real-time communication prototype that combines gesture detection with a 3D avatar is proposed in this work to facilitate conversation between hearing and deaf-mute people. In addition to employing a CNN-based VGG-16 model to recognize and interpret sign motions into text and speech, the system uses a 3D animated avatar to translate spoken or written input into Indian Sign Language (ISL). In order to improve authentic interaction, the avatar uses emotion-based face expressions and dynamically creates finger gestures. For those with hearing impairments, the prototype improves accessibility and quality of life by integrating sign-to-speech and speech-to-sign conversion modules to provide an inclusive and user-friendly communication environment.

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**Keywords:** Gesture Recognition; 3D Avatar Animation; Real-Time Avatar Interaction; Speech-to-sign translation; Convolutional Neural Network (CNN); Indian Sign Language (ISL); Avatar-Based Communication

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## 1. Introduction

Access to effective means of communication is an important human right in a world marked by linguistic diversity. Despite the fact that language is widely used for speaking and writing, a significant portion of the global population is unable to express their ideas in written form. According to the World Health Organization, 466 million people

worldwide—roughly 6% of the global population—have lost their ability to hear [1,17]. Hearing impairment has resulted in the exclusion of deaf individual from mainstream society and closure for them, sign language is a way of expression and complexity, an avenue for the consumption of communication beyond the loss of hearing. Traditionally, mediators have overcome the barrier to getting messages across to a wider and more receptive public. Individuals proficient in sign language and spoken or written language. Although these interpreters play an important role in facilitating communication, relying on third parties results in complexity and inefficiency [2,15].

In many circumstances, the absence of a sign language interpreter can isolate individuals who are deaf individual from fully participating in educational, professional, social, and personal interactions. According to the 2018 census, 63 million people, or around 6% of India's total population, have significant hearing loss [2, 18]. We might be able to write letters, phrases, and even sentences in regular spoken language using sign language, which combines hand gestures and signs. Such communication can be used by someone who has a significant hearing loss to speak or interact with others. These technologies are serving as a conduit for communication or a link between the deaf individual and the general public. With video conferencing, everyone should be able to participate, including users who use sign language [3, 16]. The majority of video conferencing systems concentrate the window on people who speak loudly, making it difficult for signers to have the freedom or chance to speak in front of a group and to do so effectively. Video conferencing presents a challenge for real-time sign language recognition since applications must do classification utilizing a large amount of stream of video as inputs, which is a logically costly operation [3,19]. Few studies have been conducted on the identification of sign language because of these difficulties. With the integration of 3D pose estimation, realistic sign language animation, and an intuitive user interface, this comprehensive solution to bridge the interaction gap between non-signers and sign language users is offered by this initiative [4].

As computer vision and artificial intelligence advance, more advancements in precision and accessibility for sign language conversion are anticipated. In the end, this would help create a more accessible and linked world, which would be a significant step in removing obstacles to communication and promoting equal opportunities for the deaf individual [5,20]. This research suggests a method that shows deaf individual intelligible sign language via a mobile application using a 3D avatar. Further the paper focus on the generation of following modules: i) Avatar creation: The 3D avatar is created using the platform “Blender” then it is trained and animated using the video-based datasets. ii) Conversion of speech into sign: The system captures the spoken conversation of user and convert it into the text. The word correspond to the text is already mapped with the list of animation from the database, thus converting the speech into sign language using the 3D avatar. iii) Conversion of sign language into speech and text: The signs composed by the mute is transcribed into text and the text is converted into the respective speech. iv) integration to video call: Finally, the system is integrated with all the modules and the real time conversion sign language is implemented in the video calls.

The rest of this paper is organized as follow as: Section II examines the current methods, tools, and issues in 3D avatar animation and sign language translation, offering a basis for enhancing accessibility and leading future developments. Section III introduced the suggested techniques that integrate gesture recognition, emotion detection, and 3D avatar movement for smooth speech-to-sign and sign-to-speech translation, allowing for real-time bidirectional communication in video conferences. As seen in Section IV, the system produces coherent 3D avatar animations that provide natural and efficient conversation while accurately translating ISL to English and vice versa.

## II. Related works

Ribeiro et al. [6] stated that In order for a deaf individual to interpret signs, suggest a system that uses motion capture devices to record them and creates the necessary animations, which are displayed as a 3D avatar. It is a more attractive 3D avatar, with a face that has more animated features than the fake avatar that was used. It also has the ability to handle additional animations, which should eventually expand to thousands of animations that reflect LGP indications. The requirements can be met by motion capture (MoCap)-based 3D avatar animation. As a result, we created SignCom, a concatenative synthesis system that generates eye gaze, facial expressions, and hand and body movements all at once. This technique makes it easier to generate new words, cut, paste, mix, and interpolate certain motion parts, then reassemble them to automatically create a signing avatar's 3D animation [7]. The development of the glosses database, which may contain many indicators, is also necessary. Using speech recognition technology, Eltanani, R. [8] created an application for smart mobile devices that translates Saudi speech to Saudi Sign Language.

A small amount of face movement mimicry was seen; however, emotional facial mimicry was not evoked by avatars. Another finding is that the ability to identify emotions is negatively impacted by the avatars' increased realism [9]. In order to generate the appropriate SSL, the application analyses the sign language while viewing the video and features a 3D avatar. The ultimate goal is to deploy a mobile SSL application that directs users—both individuals and businesses—as well as organizations toward investment possibilities across many industries.

Younes, O. [10] A useful tool for improving sign language communication involves a web-based sign language Animator (SLA) platform created in this study, which is accessible to the general public at SLAnimator.vercel.app. An overview of the project is provided in [11], which then goes on to discuss the development of an English-to-sign language translation system and the creation of an XML-compliant notation for deaf and dumb individual (SiGML) sign language gestures that are used as input to the signing avatar. Human involvement was required for the semantic and syntactic processing after the automatic method fails in the semi-automated translation from the source language, English, to the destination language, sign. This is mainly due to sign languages having their own phonology, morphology, and syntax [12]. According to Rekha et al. [13], an ISL-based handheld interpreter uses a voice input via a mobile device to generate a text message that is translated into a signed language and saved in a cloud database for future sign language retrieval. Using personalized CNNs faces several potential challenges that need to be carefully considered. Relying only on video data for the learning approach can potentially introduce inconsistent lighting or motion blur which can significantly affect the precision of facial tracking. In [14] Waldow, K et al. have developed a method that can easily extend existing facial recognition systems using personalized CNNs to provide more authentic avatar-based sign language communication.

## 2. Proposed System

This proposed system uses a 3D avatar in video conferences to translate spoken words into Indian Sign words and vice versa, so enabling real-time interaction between the hearing and deaf communities. Fig.1. Outlines the process of integrating the sign to speech/text conversion module and speech to sign/text conversion module in video call in modular way. This method incorporates gesture recognition, emotion detection, and avatar animation to ensure this interactive and inclusive kind of communication environment.

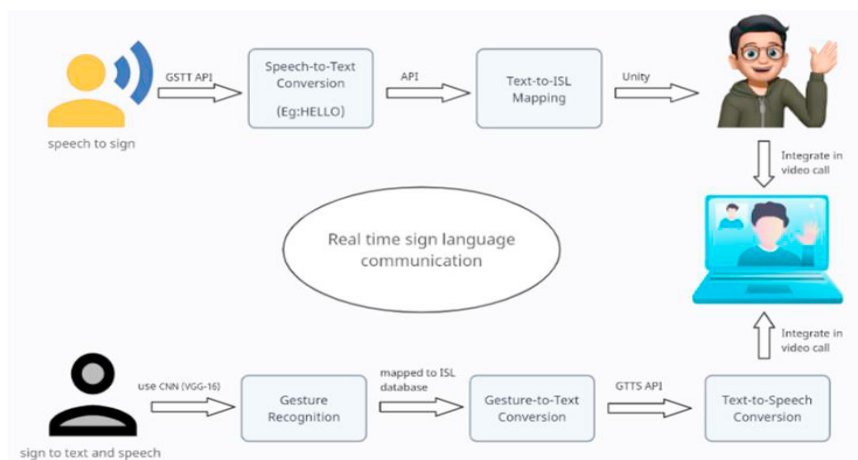


Fig. 1. Flow Diagram of Real time sign language communication

### 3.1 Avatar Animation

The model of the 3D avatar is created using the software like Blender or Unity that models human character, hand and body movements to life. To get the exact human like representations, inputs through the user are synced with the

avatar's movements through motion tracking and thus display very fluid and smooth gesture animation. The 3D model that are pre-built and stored in open-source platform can also be used thus resulting in the minimized time and effort. The process to give the animation to avatar can be classified as:

- Avatar Model creation
- Motion capture
- Animating using blender

#### a. Avatar Model Creation

Using 3D modeling software like Blender or Autodesk Maya, a 3D mesh based on the design is created. The model's geometry should be detailed, with attention to skin, clothing, and accessories, which will make the avatar more realistic. After the model is created, rigging is performed, where a skeleton structure (bones and joints) is added to the avatar. This allows for animation and movement of different body parts. The avatar's surface is then textured using UV mapping techniques, where colors, patterns, and materials (such as skin, hair, and fabric) are applied to the 3D model. This can also be achieved using the open-source platform such as Mixamo, Open3DModel, Sketchfab that allow us to download and use the pre-built 3D characters in .fbx format.

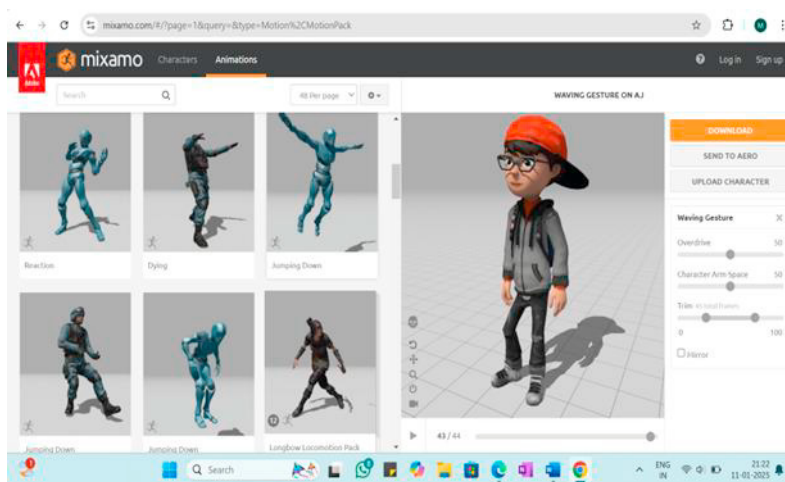


Fig. 2. Pre-built 3D-Avatar model from mixamo

Figure.2 Shows the pre-built 3D-Avatar model available in the .fbx , .glb and many other formats on the free open-source platform like mixamo from where the models can be downloaded without creating it from the beginning

#### b. Motion Capturing

The motion-capture process uses Rokoko, an application that allows for the capturing movements of a 3D avatar in real time. High-precision motion capture from Smartsuit Pro by Rokoko captures each body movement for digital animation in real time. In order to preserve the animation effects, constraints-based movement retargeting first alters initial 3D motion information before applying it to new characters under certain restrictions. This is crucial for computer animation. The full-body movements are tracked within the suit in terms of its joints, with realistic gesture expressions for sign language. In addition, Rokoko Face Capture also captures facial expression using a smartphone that will be synchronized into the 3D avatar. All these features can help in the dynamic replication of both body gestures and emotions. The motion data captured is imported into 3D software, such as Blender, to fine-tune its motion and integrate it with the avatar, and Unity. The tools of Rokoko ensure the smooth, natural movements of the avatar in close alignment with real-world actions, thus increasing the inclusive-ness and accuracy of the system.



Fig. 3. Motion capturing using recorded sign video in Rokoko studio

Figure.3 Shows the motion capturing of pre-recorded sign video where the body movements are captured and given to the skeletal structure which is then used to animate the 3D model.

### c. Animating using blender

Since virtual character animations are more efficient than traditional manual animations, which take a lot of time and money to create, they have recently become more and more popular in both the industrial and research sectors. Animating 3D models in Blender using Rokoko streamlines a workflow for making the motion very realistic. Import the rigged 3D model into Blender, set it up, and then proceed with animation. Using the tools such as Smartsuit Pro from Rokoko, one can capture the motion, and import this into Blender via Rokoko Studio Live plugin. This plugin allows seamless retargeting of the captured motion data onto the 3D model. In Blender, the Retargeting Process aligns the rig structure of the motion data with the model's skeleton so that the movement is mapped properly. The animation is fine-tuned in Pose Mode in Blender, where discrepancies such as misaligned joints or exaggerated movements are corrected. Refining motion curves, optimizing timing, and smoothing transitions are accomplished using the Graph Editor and Dope Sheet. Once the animation is finalized, it may be exported for real-time applications or can be integrated into the project and the avatar can perform lifelike gestures and actions. Blender combined with Rokoko provides an effective and accurate system for the animation of the 3D avatar.

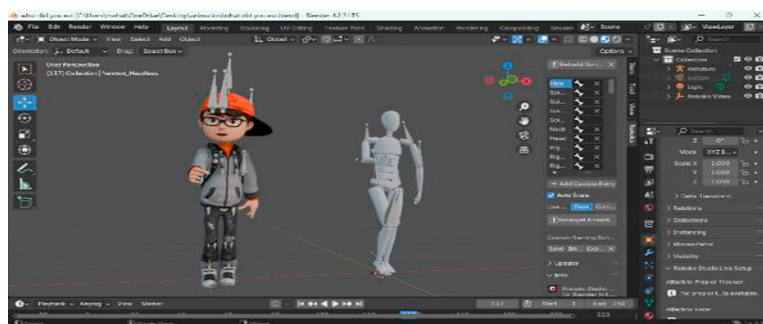


Fig. 4. Animating 3D Avatar model in Blender platform

Figure.4. Illustrates animating the avatar model by integrating the motion captured using the rokoko studio which can also be used from the blender platform by retargeting animations. The applied corresponding keyframe animations model each sign, and movements are controlled with the application of inverse kinematics that account for bone rotations and joint translations. Real-time recognition of facial expressions further enriches the avatar animations that make communication emotionally engaging and natural for the user. The transformation matrix  $T$  is expressed as follows where the  $T(dx,dy,dz)$  is translation, ensures that each sign is executed correctly.

$$T=R(\theta)\times T(dx,dy,dz) \quad (1)$$

Where;  $R(\theta) = \text{rotation}$

$T(dx,dy,dz) = \text{translation}$

### 3.2 Sign-to-Speech and Text Conversion

The conversion of sign language into voice and text depends on gesture recognition. A deep learning model built on the architecture of a VGG-16 convolutional neural network, or CNN, is used to do this. The pre-trained deep learning model VGG-16 will be used to identify sign language motions from picture or video inputs. First, this CNN model processes video feeds of the user's hand movements. Frame extraction will take place in which each frame would undergo preprocessing stages, including resizing and normalization to have the dimensions aligned with VGG-16's input dimensions. Frames are processed by the CNN, which extracts spatial features and classifies them into their corresponding sign language gestures. These are mapped into specific phrases or words in the system. The moment a gesture is recognized is when the corresponding text will be displayed as output. The CNN model classifies these gestures into specific ISL signs and translates them into corresponding text and speech. The formula behind this process involves convolution operations on the video frames, represented as

$$(I*k)(x,y)=\sum_m \sum_n I(m,n).K(x-m,y-n) \quad (2)$$

Where;  $I$  = input frame,

$K$  = convolution kernel

Once the gesture is categorized, the system has a set of predefined mappings that it uses to convert this gesture into text and produces speech output by using text-to-speech APIs, therefore, making communication coming from a deaf individual understandable to a hearing person. The GTTS API translates the recognized text into speech for speech conversion. The text is passed to GTTS, creating a speech file in real time. This way, there will be an auditory output apart from the display of the text. For the purpose of inclusive communication, this integrated method makes it fast and accessible for everyone to translate sign language into speech and text.

### 3.3 Speech-to-Sign Translation

To enable hearing persons to communicate with deaf individual, the system includes a speech-to-sign module that converts spoken words into ISL gestures by making use of the 3D avatar. Signing avatars have the potential to make sign languages more accessible and valuable in everyday life, which can help overcome the challenges faced by the Deaf community when it comes to accessing information. The speech-to-sign/text conversion system would use the Text-to-Speech (GTTS) API from Google for speech recognition that transcribes spoken words into text and the generated avatar to map out sign animation [15]. The spoken input would first be recorded by a microphone, and the audio input would be sent into GTTS to analyse the voice in converting to text. This text then acts as the intermediary format to map speech into corresponding signs. The recognized text will be mapped on predefined animations on the avatar about sign language gestures. It appears that this method of assembling sign subunits to produce animated motions is effective. As more dynamic movements are specified, the approach will be improved. There is a relationship between each word or phrase, and specific animations created using Blender to ensure precise representation of its meaning. They are rendered live through the avatar, allowing it to give a visual for the content that is verbally spoken. Our goal is to use a 3D signing avatar to develop a new tool for Sign Writing that synthesizes the movements expressed within a written form of sign language. Thus, this system allows for an easy conversion from spoken language to sign language to text, so there is actually a bi-directional communication bridge of deaf individual and hearing individuals. GTTS for speech recognition and the avatar for sign representation ensure accessibility in real-time conversations.

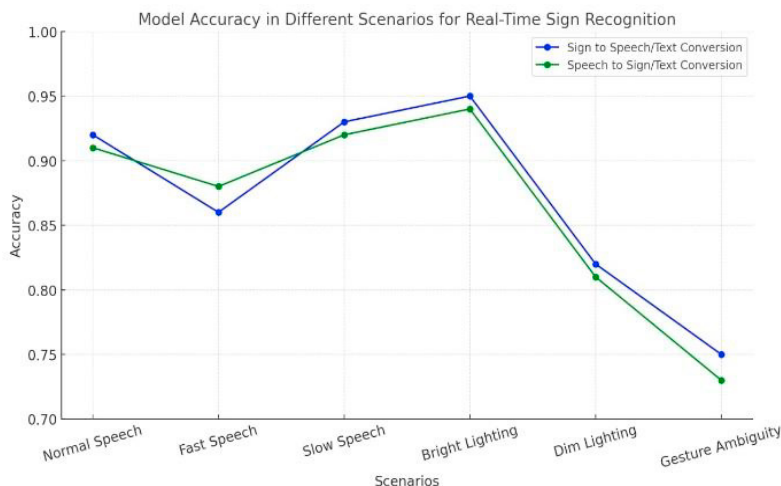


Fig. 5. Accuracy graph plotted based on different scenarios.

Figure.5. Illustrates the accuracy obtained from the implementation of this system based on different scenarios such as ‘speech’ (Fast, Normal, Slow), ‘Lighting conditions’ (Bright, Dim) and the Gesture Ambiguity.

### 3.4 Integration in Video Calls

WebRTC, which enables peer-to-peer, real-time communication via web browsers, is one of the most widely used technologies for video calling. Finally, all the modules will be deployed in the video call system, such as WebRTC, and ensure that the avatar has real-time translation and gesture recognition while engaged in conversation. WebRTC also manages the data and connections for video chats. The video call interface, developed in frameworks like Flutter, will facilitate a seamless user experience and has provision for speech-to-sign and sign-to-speech switching modes. The system should be adjusted to speech, gestures, and facial expressions, with real-time synchronizations. Then, it has to respond in such a way that its movements and expressions will suit the conversation that is going on and should not happen with palpable latency. Thereby, minimizing latency, smooth transitions between the recognized gestures, and speech must be put into place.

## 3. Result and Discussion

50 frequently used signs (such as I, sorry, thank you, please, welcome, good morning, good evening, good night, today, tomorrow, yesterday) and user-to-user conversations are included in this extensive sign language database. The dataset covers the most often used keywords in sign language and consists of 78 unique sentences with 324 different words. To ensure correct depiction, the Blender toolset was used to create and animate the sign movements that corresponded to each word. We created four animation sequences each phrase to improve clarity by capturing motion variances. To further illustrate key sign movements in a simplified way, we offer sample animation sequences.

This section defines the steps involved for translating ISL into its English equivalent. It has been demonstrated that the sign movement of the words in an English phrase correspond to each other. To translate English to sign movements, the Natural Language Toolkit (NLTK) is used. An example plot showing how a ISL is changed from an English sentence is given. As illustrated in Fig. 6, the individual exhibits sign language, which is subsequently translated into the corresponding English expression: "I'm fine." The system's translation mechanism, which transforms input speech into text and creates dynamic 3D signing avatars in response, is shown in Fig. 7. These avatars precisely represent the sign language motions connected to the identified dialog and are smoothly superimposed on the screen.



Fig. 6. Output depicting the sign to speech/text translation.

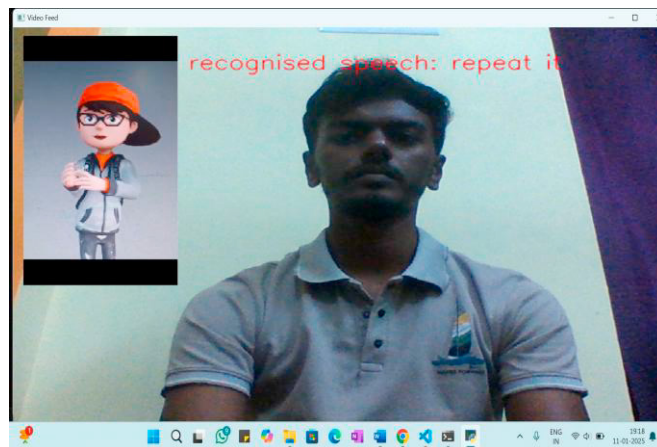


Fig. 7. Output depicting the speech to sign/text translation

## 5 Conclusion

This system has bridged the gap between deaf individuals and hearing individuals by employing state-of-the-art technology devices, including 3D avatar animation, gesture recognition, spoken words to sign translation and sign to voice conversion. Hence, using CNN-based VGG-16 deep learning models along with WebRTC frameworks for real-time work is an excellent medium of communication seamlessly and immersively within the real-time conversation via a video call. Thus, the system ensures that both the hearing and the deaf individuals communicate effectively with no barrier as it translates spoken words into Indian Sign Language gestures using an avatar; then the gestures back to text and speech. In addition, the Inclusion of facial emotion recognition has enhanced the user experience since the avatar interacts on human-to-human scale, acting more human-like, with emotional expressions. With real-time ease of interaction and inclusiveness, this approach promises to take leaps for enhancing communication for deaf individuals across the domains of education, health sectors, and social platforms. This finally demonstrates how technology can be put to use for bringing greater accessibility and inclusiveness in human communication.

## References

- [1] Davey, S., Maheshwari, C., Raghav, S.K., Singh, N., Muzammil, K. and Pandey, P., (2018). Impact of Indian public health standards for rural health care facilities on national programme for control of deafness in India: The results of a cohort study. *Journal of family medicine and primary care*, 7(4), pp.780-786.
- [2] Patel, B.D., Patel, H.B., Khanvilkar, M.A., Patel, N.R. and Akilan, T., (2020), August. ES2ISL: an advancement in speech to sign language translation using 3D avatar animator. In *2020 IEEE Canadian Conference on Electrical and Computer Engineering (CCECE)* (pp. 1-5). IEEE.

- [3] Gupta, V., Punjani, R., Vaswani, M. and Kundale, J., (2022), August. Video Conferencing with Sign language Detection. In *2022 2nd Asian Conference on Innovation in Technology (ASIANCON)* (pp. 1-5). IEEE.
- [4] Pawar, S., Shastri, Y., & Aiman, S. Z. (2024, March). Bidirectional Sign Language Assistant with MediaPipe Integration. In *2024 International Conference on Emerging Smart Computing and Informatics (ESCI)* (pp. 1-8). IEEE.
- [5] Reddy, S.B., Shahebaaz, A., Dodda, R. and Raghavendra, C., (2024), March. Transformative Advancements: Sign Language Conversion to Text and Speech. In *2024 International Conference on Emerging Smart Computing and Informatics (ESCI)* (pp. 1-7). IEEE.
- [6] Ribeiro, B., Dias, D., Faria, P.M. and Romero, L., (2023), June. Capturing and Processing Sign Animations to a Portuguese Sign Language 3D Avatar. In *2023 30th International Conference on Systems, Signals and Image Processing (IWSSIP)* (pp. 1-5). IEEE.
- [7] Furqatovich, Islombek Umirov, Rabbimov Mukhriddin Musoqulo'li, Usmanova Dilafruz Karshievna, Christo Ananth, and T. Ananth Kumar. (2024)"Artificial Intelligence Powered Strategies to Enhance the Competitiveness of Healthcare Institutions in Uzbekistan." In *2024 Ninth International Conference on Science Technology Engineering and Mathematics (ICONSTEM)*, pp. 1-6. IEEE.
- [8] Eltanani, R., Alwadia, K., Altuwayjiri, G. and Al-Shargabi, A.A., (2023), October. SawtiEshara: An Avatar-Based Translator from Saudi Speech to Saudi SignLanguage. In *2023 3rd International Conference on Emerging Smart Technologies and Applications (eSmarTA)* (pp. 1-8). IEEE.
- [9] Saquinaula, A., Juarez, A., Geigel, J., Bailey, R. and Alm, C.O., (2022), March. Emotional Empathy and Facial Mimicry of Avatar Faces. In *2022 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)* (pp. 770-771). IEEE.
- [10] Younes, O., (2024), Sign Language Animator: A Platform for Multilingual Sign Language Synthesis. In *2024 International Conference on Intelligent Systems and Computer Vision (ISCV)* (pp. 1-8). IEEE.
- [11] Dhanjal, A. S., & Singh, W. (2025). Multilingual speech to Indian sign language translation using synthetic animation: a resource-efficient approach. *Multimedia Tools and Applications*, 1-33. Sutton-Spence, R. and Woll, B., 1999. *The linguistics of British Sign Language: an introduction*. Cambridge University Press.
- [12] Dwivedi, J. N., Dhabuwala, K., Padhara, R., Kulkarni, P., & Patel, D. (2025, March). Gesture Language Translation Using Convolutional Neural Network. In *2025 International Conference on Emerging Smart Computing and Informatics (ESCI)* (pp. 1-5). IEEE.
- [13] Waldow, K., Fuhrmann, A. and Roth, D., (2024), March. Facial Feature Enhancement for Immersive Real-Time Avatar-Based Sign Language Communication Using Personalized CNNs. In *2024 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)* (pp. 919-920). IEEE.
- [14] Ajagbe, S. A. (2024). Developing Nigeria Multilingual Languages Speech Datasets for Antenatal Orientation. In H. Florez, & H. (. Astudillo (Ed.), *Applied Informatics. ICAI 2024. Communications in Computer and Information Science*. 2237, pp. 157-170. Springer, Cham.
- [15] Adeniyi, J. K., Ajagbe, S. A., Adeniyi, A. E., Aworinde, H. O., Falola, P. B., & Adigun, M. O. (2024). EASESUM: an online abstractive and extractive text summarizer using deep learning technique. *IAES International Journal of Artificial Intelligence (IJ-AI)*, 13(2), 1888-1899. Retrieved from <https://ijai.iaescore.com/index.php/IJAI/article/view/23221>.
- [16] Najib, F. M. (2025). Sign language interpretation using machine learning and artificial intelligence. *Neural Computing and Applications*, 37(2), 841-857.
- [17] Mosa, D. T., Nasef, N. A., Lotfy, M. A., Abohany, A. A., Essa, R. M., & Salem, A. (2023). A real-time Arabic avatar for deaf-mute community using attention mechanism. *Neural Computing and Applications*, 35(29), 21709-21723.
- [18] Chen, S., Cheng, H., Su, S., Patterson, S., Kushalnagar, R., Huang, Y., & Wang, Q. (2025). Customizing Generated Signs and Voices of AI Avatars: Deaf-Centric Mixed-Reality Design for Deaf-Hearing Communication. *Proceedings of the ACM on Human-Computer Interaction*, 9(2), 1-31.
- [19] Vikranthnivas, D., Venkatesh, P., Narendran, M., & Anju, A. (2025, June). AVATASIGN: Enhancing Sign Language Communication with AI powered User Customized Avatars. In *2025 3rd International Conference on Self Sustainable Artificial Intelligence Systems (ICSSAS)* (pp. 1146-1152). IEEE.